

Climbing the Plant Ladder

I PUC · Ch 3 · Plant Kingdom

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



The Green Ladder

Prof. BAIO drew a tall green ladder. "Plants climb from simple to complex: **algae**, **bryophytes**, **pteridophytes**, **gymnosperms** and finally **angiosperms**." Bindu stepped on the first rung — and splashed straight into water!

💡 **REMEMBER**

Plant groups in order of complexity: Algae → Bryophytes → Pteridophytes → Gymnosperms → Angiosperms.



The Algae Colour Party

Three algae families were having a party. The green **Chlorophyceae** stored starch. The brown **Phaeophyceae** stored laminarin and mannitol and wore fucoxanthin. The red **Rhodophyceae** showed off r-phycoerythrin and stored **floridean starch**.

💡 **REMEMBER**

Green: chlorophyll a, b · Brown: chlorophyll a, c + fucoxanthin · Red: chlorophyll a, d + r-phycoerythrin.



Algae at Work

"We are useful!" said the algae. "Half of all the carbon fixed on Earth is fixed by us." **Agar** from Gelidium helps scientists grow microbes. Protein-rich **Chlorella** is food even for space travellers. Algin and carrageen make ice-cream smooth.

💡 **REMEMBER**

Agar — Gelidium, Gracilaria · Algin — brown algae · Carrageen — red algae · Chlorella — protein-rich food.



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Amphibians of the Plants

On the second rung lived mosses and liverworts — the **bryophytes**. "We live on land but need water to reproduce, so we are the amphibians of the plant kingdom." Their main body was the **gametophyte**. **Sphagnum** proudly said, "I give peat, a fuel!"

💡 REMEMBER

In bryophytes the gametophyte is dominant and the sporophyte depends on it. Marchantia is a liverwort; Funaria is a moss.

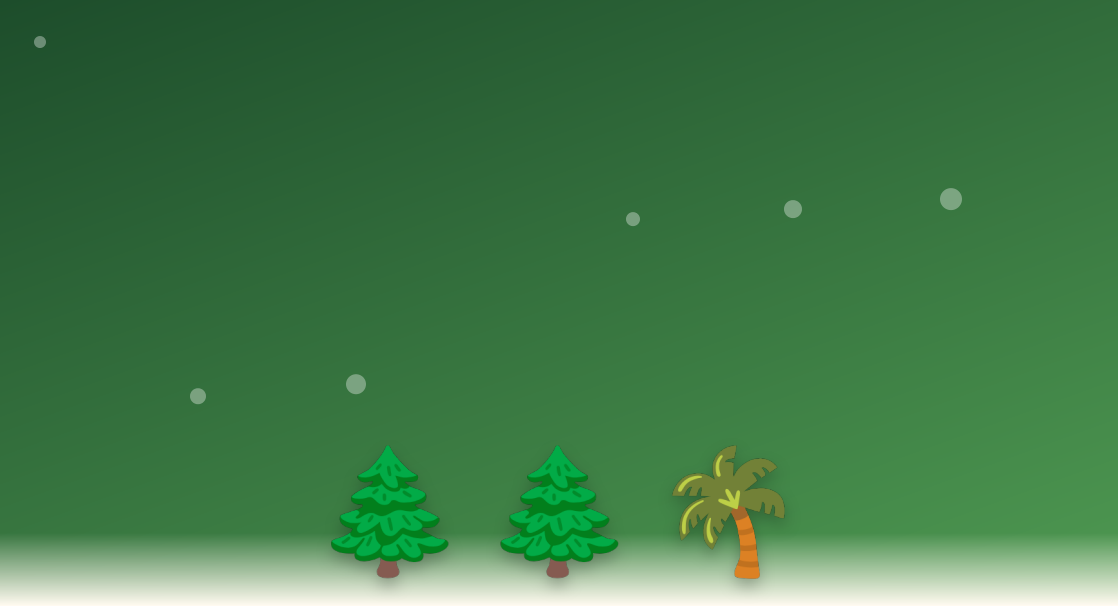


The First Pipes

Ferns on the third rung showed their pipes — **xylem** and **phloem**. "We are the first land plants with vascular tissue!" Here the main plant was the **sporophyte**. Selaginella and Salvinia made two kinds of spores — they were **heterosporous**.

💡 **REMEMBER**

Pteridophytes: the first terrestrial plants with xylem and phloem. Heterospory is a step towards the seed habit.



Naked Seeds

Tall trees stood on the fourth rung — the **gymnosperms**. "Our ovules are not covered by an ovary, so our seeds are naked." **Cycas** had **coralloid roots** with nitrogen-fixing cyanobacteria. Pinus had fungal partners called **mycorrhiza**. Sequoia stood among the tallest trees on Earth.

💡 **REMEMBER**

Gymnosperms have naked seeds. Cycas — coralloid roots · Pinus — mycorrhiza, needle leaves · Sequoia — very tall.



Fruits at the Top

At the very top, flowers bloomed — the **angiosperms**. "Our seeds are safely inside fruits!" Tiny **Wolffia** waved from a pond, and **Eucalyptus** touched the clouds. Their special trick was **double fertilisation**.

💡 REMEMBER

Angiosperms: seeds enclosed in fruits. Wolffia is the smallest; Eucalyptus grows over 100 m tall.



Round and Round

Prof. BAIO summed up the life cycles. "When the haploid gametophyte dominates, as in many algae, the cycle is **haplontic**. When the diploid sporophyte dominates, as in seed plants, it is **diplontic**. Bryophytes and pteridophytes share both — they are **haplo-diplontic**."

💡 **REMEMBER**

Haplontic — many algae · Diplontic — gymnosperms and angiosperms · Haplo-diplontic — bryophytes, pteridophytes.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

algae p.1 · bryophytes p.1 · pteridophytes p.1 · gymnosperms p.1 · angiosperms p.1 · Chlorophyceae p.2 · Phaeophyceae p.2 · Rhodophyceae p.2 · floridean starch p.2 · Agar p.3 · Chlorella p.3 · gametophyte p.4 · Sphagnum p.4 · xylem p.5 · phloem p.5 · sporophyte p.5 · heterosporous p.5 · Cycas p.6 · coralloid roots p.6 · mycorrhiza p.6 · Wolffia p.7 · Eucalyptus p.7 · double fertilisation p.7 · haplontic p.8 · diplontic p.8 · haplo-diplontic p.8



QUICK QUIZ

Do you remember?

1. The "amphibians of the plant kingdom" are:

- Algae
- Bryophytes
- Pteridophytes
- Gymnosperms

2. Agar is obtained from:

- Gelidium
- Laminaria
- Chlorella
- Spirogyra

3. The first land plants with vascular tissue are:

- Bryophytes
- Pteridophytes
- Algae
- Fungi

4. Coralloid roots are found in:

- Pinus
- Cycas
- Wolffia
- Funaria

5. Red algae store food as:

- Starch
- Laminarin
- Floridean starch
- Glycogen

Answers: 1-Bryophytes · 2-Gelidium · 3-Pteridophytes · 4-Cycas · 5-Floridean starch